



The Superior University, Lahore

LAB-TASK (11) Spring-2026

Course Title:	Programming for AI			Course Code:	-	Credit Hours	1
Instructor:	Sir rasikh			Program:	BSDS		
Semester:	4th	Batch:	Morning	Section:	4A	Date:	-
Time Allowed:	-			Maximum Marks:	N/A		
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Work Assigned:

Question:

What is the LangChain, LLMs, RAG, FAISS, Vector, VectorDB, GenAI, and GANs? Explain each concept briefly with examples.



1. LangChain

LangChain is a framework that is used to make applications with LLMs. It is not a model itself, but it helps to connect language models with different things like documents, APIs, databases, tools and memory.

In simple words, LangChain works like a bridge between an AI model and other external sources. If we want to make a chatbot that can answer from PDF files or university documents, then LangChain can help us to connect all these things together.

For example, if we make a university admission chatbot, the LLM alone may not know the latest admission dates or fee structure. With LangChain, we can connect the chatbot with admission files, database, or website data. Then the chatbot can give better and useful answer.

LangChain is mostly used in AI chatbots, document question answering systems, personal assistants, and customer support bots.

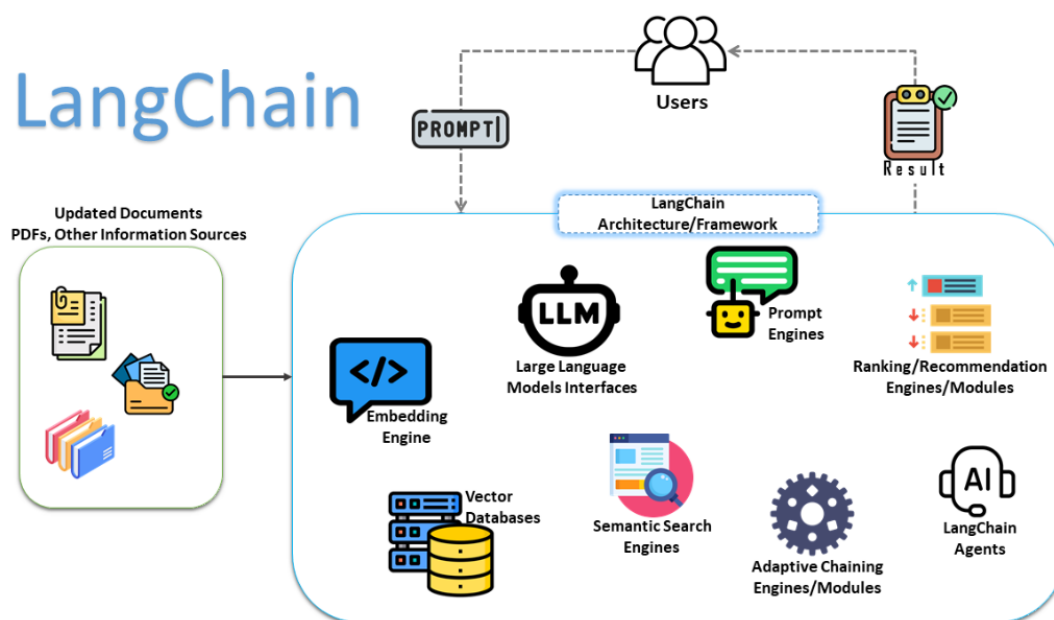
Example:

A student asks: “What is the admission deadline?”

LangChain can help the chatbot search the admission document and then send that information to the LLM for answer.

Usage

LangChain helps developers to build complete AI applications using LLMs.



2. LLMs (Large Language Models)

LLMs stands for Large Language Models. These are deep learning models that are trained on very large amount of text data. They can understand human language and also generate human-like text.

LLMs are used in many applications like ChatGPT, chatbots, translation tools, summarization tools, coding assistants and virtual assistants. These models learn grammar, sentence structure, meaning of words, and patterns from huge text data.

For example, when we ask ChatGPT any question, it understands our question and generates answer in normal language. This is because ChatGPT is based on a Large Language Model.

LLMs can do many tasks such as answering questions, writing paragraphs, generating code, translating languages, summarizing long text, and helping in conversations.

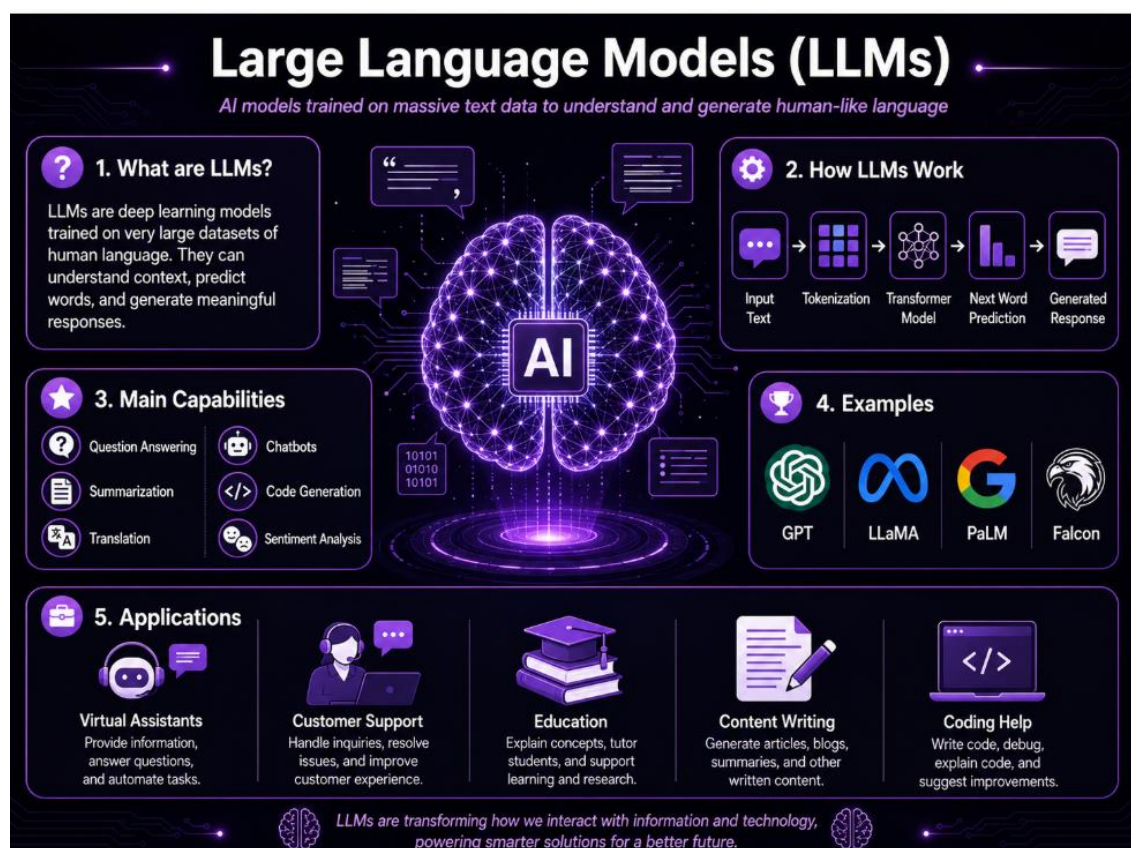
Example:

If user asks: “Explain Artificial Intelligence in easy words.”

The LLM can generate a complete answer in simple language.

Main uses of LLMs:

- Chatbots
- Text generation
- Summarization
- Translation
- Sentiment analysis
- Code generation
- Question answering



3. RAG (Retrieval-Augmented Generation)

RAG means Retrieval-Augmented Generation. It is a technique in which we combine searching with language generation.

In simple words:

RAG = Search + LLM

A normal LLM gives answer from its trained knowledge. But sometimes we need updated or private information, like university rules, hospital records, company documents, or PDF notes. LLM may not already know this data. So RAG is used.

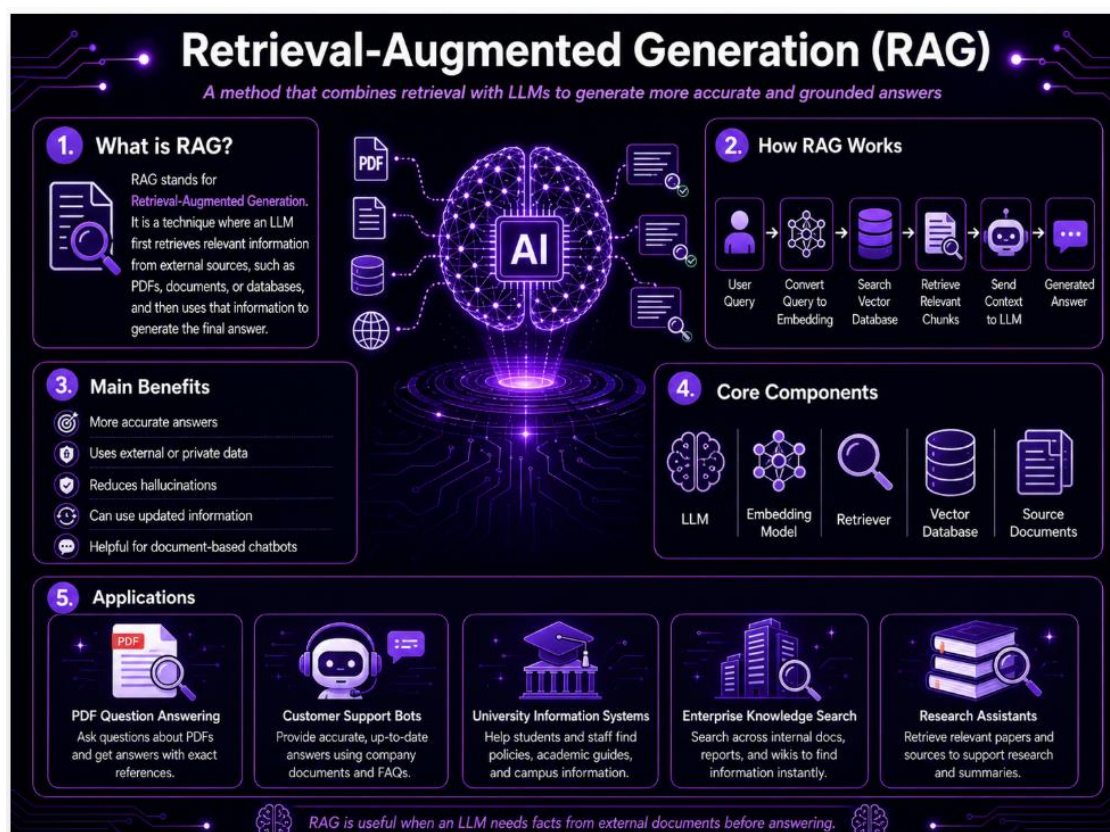
In RAG, first the system searches the relevant information from external documents or database. After that, this information is given to the LLM. Then LLM generates the final answer using that information.

For example, if a student asks, “What is the last date of admission?”, the system first searches the admission document. Then it finds the deadline and LLM writes a proper reply.

RAG is useful because it reduces wrong answers and makes chatbot more accurate. It is mostly used in document-based chatbots.

Example:

A university chatbot can use RAG to answer from admission policy, fee structure, and program files.



4. FAISS (Facebook AI Similarity Search)

FAISS stands for Facebook AI Similarity Search. It is a library made by Facebook AI. It is used for fast searching of similar vectors.

In AI, documents, words and sentences are converted into vectors. These vectors are numeric form of meaning. When user asks a question, that question is also converted into a vector. FAISS compares this question vector with stored vectors and finds the most similar one.

For example, if user asks “Tell me admission fees”, FAISS can find the document paragraph related to fee structure because both meanings are similar.

FAISS is very fast, so it is useful when we have large number of documents. It is mostly used in RAG systems, recommendation systems, semantic search, and question-answering systems.

Example:

If we have 1000 paragraphs from university documents, FAISS can quickly find the paragraph which is most related to user question.



5. Vector

A vector is a list of numbers that represents data. In AI, text, image, audio or sentence can be converted into vectors so computer can understand their meaning.

Computer cannot directly understand words like humans. So we convert words or sentences into numbers. These numbers are called vectors or embeddings.

These numbers represent the meaning of the word. Similar words have similar vectors. For example, “college” and “university” will have close vectors because their meaning is related. But “college” and “banana” will not be close because their meaning is different.

Vectors are very important in AI because they help machine understand similarity between words, sentences, images and documents.

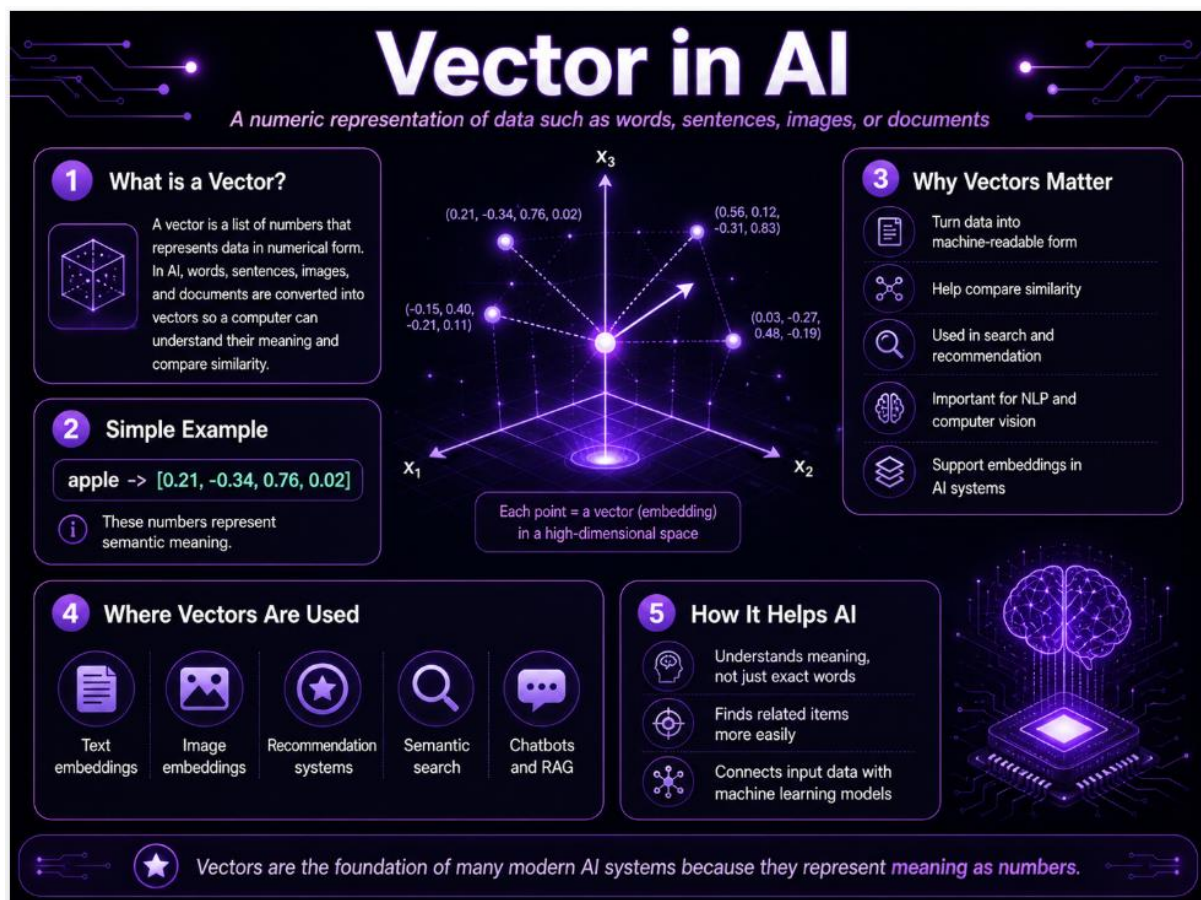
Example:

Sentence: “I lost my ID card”

This sentence can be converted into a vector, and then chatbot can match it with lost item data.

Main point:

Vector is numeric representation of data meaning.



6. VectorDB (Vector Database)

VectorDB means Vector Database. It is a special type of database that stores vectors and allows fast similarity search.

Normal database stores data like name, email, phone number, ID, marks etc. But VectorDB stores embeddings or vectors. These vectors can represent documents, questions, images, or sentences.

VectorDB is very useful in GenAI and RAG applications. When user asks a question, it is converted into vector. Then VectorDB searches which stored vector is most similar to the question. After that, related document or answer is returned.

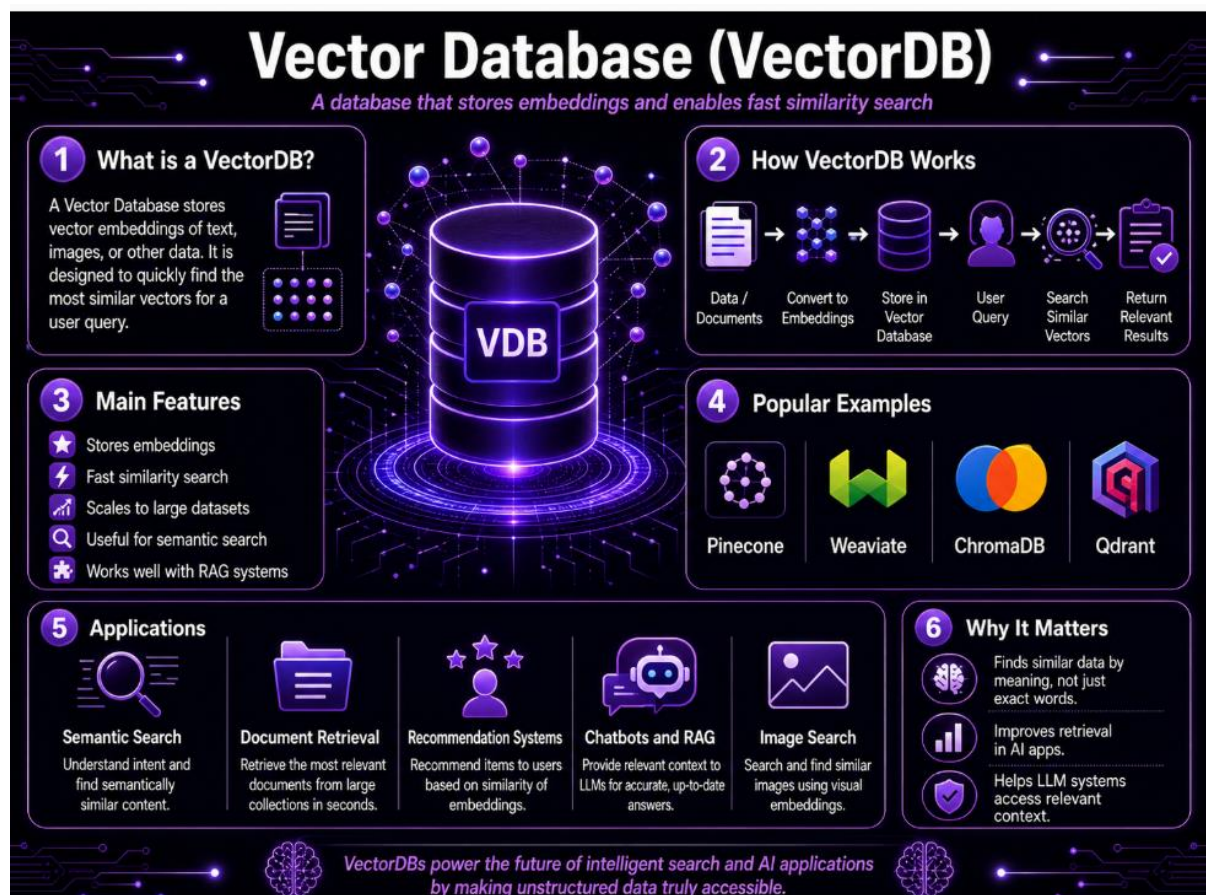
Popular VectorDB examples are Pinecone, Weaviate, ChromaDB and Qdrant.

For example, in a library assistant chatbot, all book descriptions can be stored as vectors. When student asks “Python programming book”, VectorDB can find books related to Python even if exact words are not same.

Example:

User asks: “I need book for learning coding.”

VectorDB can find “Python Programming for Beginners” because meaning is similar.



7. Generative AI (GenAI)

Generative AI, also called GenAI, is a branch of Artificial Intelligence that creates new content. It can generate text, images, music, videos, designs and code.

Traditional AI mostly classifies or predicts things. For example, it may tell whether an email is spam or not. But Generative AI creates something new. It learns patterns from data and then generates new similar content.

For example, ChatGPT generates text answers. DALL·E generates images. Codex generates code. Stable Diffusion generates images from text prompts.

GenAI is used in many fields like education, designing, programming, marketing, content writing and entertainment.

Common Generative AI models are:

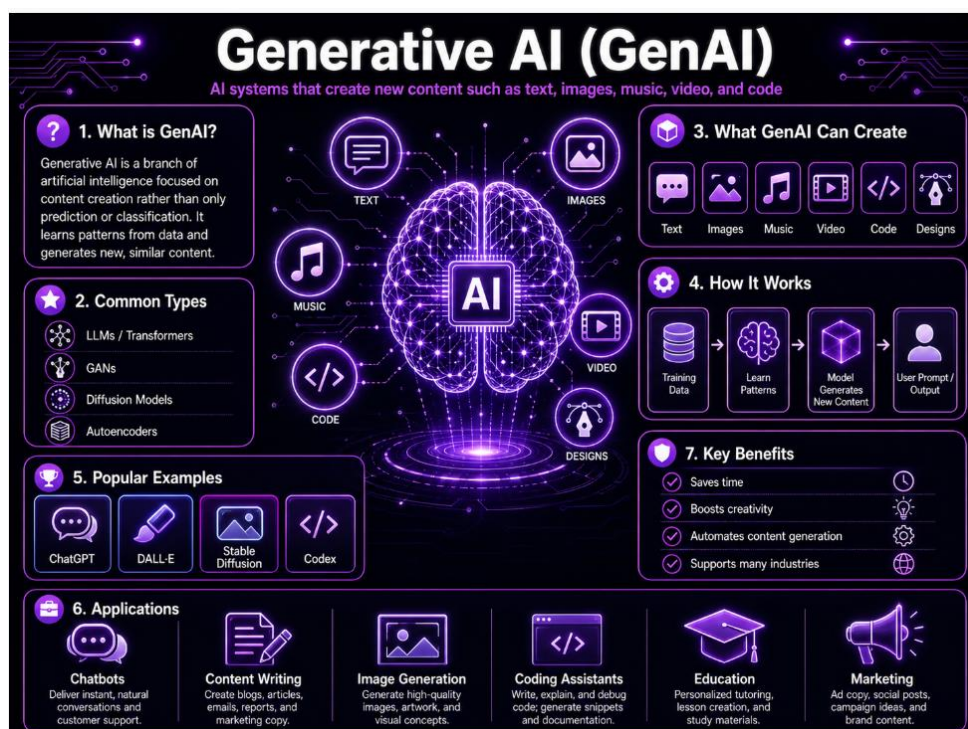
- LLMs
- GANs
- Diffusion Models
- Transformers

Example:

If user writes: “Write an essay on Artificial Intelligence”, ChatGPT can create a new essay. This is Generative AI.

Main point:

Generative AI is AI that creates new content.



8. GANs (Generative Adversarial Networks)

GANs stands for Generative Adversarial Networks. GANs are a type of Generative AI model. They are mostly used to create realistic images or synthetic data.

A GAN has two neural networks:

1. Generator

Generator creates fake data. For example, it may create a fake face image.

2. Discriminator

Discriminator checks if the data is real or fake.

Both networks compete with each other. Generator tries to make fake data more realistic. Discriminator tries to catch fake data. Slowly the generator becomes better and starts creating very realistic outputs.

For example, GANs can generate human faces that look real but those people do not actually exist.

GANs are used in image generation, image enhancement, style transfer, face generation, and data augmentation.

Example:

A GAN can generate realistic face images, cartoon images, or improved quality images.

Main point:

GANs generate realistic data by using two competing neural networks.

